CHAPTER 4

# Differentiation, convexity and continuity

S29, S31, S32; revisited in S33 and S40

**Objectives**

Differentiate a composite function; study variations and convexity; prove existence and uniqueness; approximate a solution.

## Course notes

### Differentiating a composite function

If $u$ is differentiable on $I$ and $v$ is differentiable on an interval containing $u(I)$, then $(v \circ u)' = (v' \circ u)u'$. The domains must be checked before the calculation.

$$(u^n)' = nu'u^{n-1}, \quad (e^u)' = u'e^u, \quad \left(\frac{1}{u}\right)' = -\frac{u'}{u^2}, \quad (\sqrt{u})' = \frac{u'}{2\sqrt{u}} \quad (u > 0).$$

For example, $f(x) = \sqrt{3x+1}$ is differentiable for $x > -\frac{1}{3}$ and $f'(x) = \frac{3}{2\sqrt{3x+1}}$. The sign of $f'$ determines the variations of $f$; the sign of $f$ does not.

### Convexity

A function $f$ is convex on an interval $I$ if its curve lies below its chords: for $x, y \in I$ and $t \in [0,1]$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

For a twice-differentiable function, the following equivalences are accepted: $f$ is convex; $f'$ is increasing; $f'' \geq 0$; the curve lies above all its tangents. The inequalities are reversed for a concave function. A point of inflection is a point where the curve changes convexity.

**Proof: position relative to the tangents.** Suppose $f'' \geq 0$. Fix $a$ and set $g(x) = f(x) - f(a) - f'(a)(x-a)$. Since $f'$ is increasing, $g'(x) = f'(x) - f'(a) \leq 0$ for $x \leq a$ and $\geq 0$ for $x \geq a$. Thus $g$ attains its minimum $g(a) = 0$ at $a$, so $f(x) \geq f(a) + f'(a)(x-a)$.

**Examples.** $x^4$ is convex and $f''(0) = 0$, but 0 is not the abscissa of a point of inflection. $x^3$ changes convexity at 0. The convexity of the exponential and its tangent at 0 give $e^x \geq 1 + x$.

### Continuity and equations

$f$ is continuous at $a$ if $\lim_{x \to a} f(x) = f(a)$. Every differentiable function is continuous. The converse is false: $|x|$ is continuous at 0 but not differentiable at 0.

**Intermediate value theorem**

If $f$ is continuous on $[a,b]$, every real number between $f(a)$ and $f(b)$ has at least one preimage in $[a,b]$. If $f$ is moreover strictly monotonic, this preimage is unique. One may also work on an open interval using the limits at the endpoints.

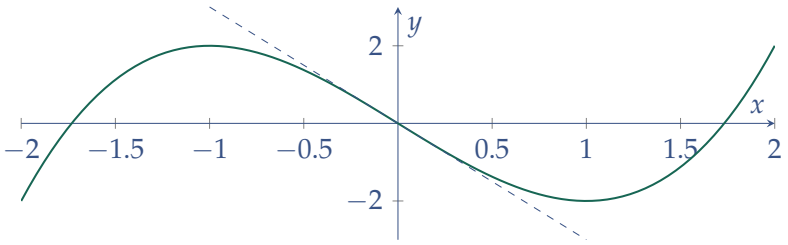
**Worked example.** $f(x) = x^3 + x - 1$ is continuous on $[0,1]$. Since $f(0) = -1$ and $f(1) = 1$, the IVT gives a zero in $]0,1[$. Since $f'(x) = 3x^2 + 1 > 0$, it is unique. $f(0.68) < 0 < f(0.69)$ provides a bracketing, without replacing the justifications of existence and uniqueness.

**A piecewise-defined function.** For $x \neq 1$, set $g(x) = \frac{x^2-1}{x-1}$, and set $g(1) = a$. Since $g(x) = x + 1$ when $x \neq 1$, $\lim_{x\to 1} g(x) = 2$. The function is continuous at 1 if and only if $a = 2$. The value at the point must therefore be compared with the limit, and not replaced by it without justification.

**Sequences and continuity.** If $u_n \to \ell$ and $f$ is continuous at $\ell$, then $f(u_n) \to f(\ell)$. For a convergent recursively defined sequence $u_{n+1} = f(u_n)$, we obtain $\ell = f(\ell)$.

### Bisection

For $f$ continuous and $f(a)f(b) < 0$, compute $m = \frac{a+b}{2}$. If $f(m) = 0$, stop; otherwise keep the half whose endpoints have images of opposite signs. After $n$ steps, the length is $\frac{b-a}{2^n}$. The midpoint gives an error of at most half the length. The method does not guarantee uniqueness; this is proved separately.




$y = x^3 - 3x$ and its tangent at the point of inflection.

## Research activity

**Derivatives and reading graphs — 35 min.**

Consider the function $f$ defined on $\mathbb{R}$ by $f(x) = x^3 - 3x$. Denote its graph by $\mathcal{C}_f$.

1. Calculate $f'(x)$ and $f''(x)$, then draw up their sign tables.
2. Deduce the variations of $f$ and its intervals of convexity and concavity.
3. Determine the coordinates of the point of inflection of $\mathcal{C}_f$ and an equation of the tangent at this point.
4. Using the limits, the extrema and the intermediate value theorem, determine the number of real solutions of $f(x) = 1$.
5. Explain, from this example, why continuity is not enough to ensure that a solution is unique.

## Progressive exercises

**E1 · Practice.** Define on $\mathbb{R}$ the functions $f(x) = (2x-1)^5$ and $g(x) = e^{-x^2}$. Calculate $f'(x)$ and $g'(x)$.

**E2 · Practice.** Let $f$ be the function defined on $\mathbb{R}$ by $f(x) = x^4 - 6x^2$. Determine the intervals on which $f$ is convex or concave.

**E3 · Practice.** Let $\mathcal{C}$ be the graph of the exponential function. Determine an equation of its tangent at the point with abscissa 1. Using convexity, deduce an inequality valid for every real number $x$.

**E4 · Practice.** Prove that the equation $x^3 + 2x = 5$ has a unique real solution. Bracket this solution between two consecutive integers.

**E5 · Practice.** A bisection algorithm starts from the interval $[0;2]$ and halves its length at each step. Determine the minimum number of steps guaranteeing an interval of length at most $10^{-3}$.

## Textbook or past-paper work

**Barbazo, p. 132, exercise 29.** In class after the study of convexity: questions 1 and 2. At home: finish questions 3 to 5. The aim is to distinguish the sign of $f'$, which governs the variations, from that of $f''$, which governs convexity.

## Homework

**Homework 4 — A maximum and an equation — 60 min.**

Consider the function $f$ defined on $[0;+\infty[$ by $f(x) = (x+1)e^{-x}$. Its graph is denoted $\mathcal{C}_f$.

1. Calculate $f(0)$ and the limit of $f$ at $+\infty$. Calculate $f'(x)$, then draw up the table of variations of $f$.
2. Calculate $f''(x)$. Study the convexity of $f$ and determine the coordinates of the possible point of inflection of $\mathcal{C}_f$.
3. Prove that the equation $f(x) = \frac{1}{2}$ has a unique solution $\alpha$ on $[0;+\infty[$.
4. Write a Python function that uses bisection on $[0;2]$ and returns two real numbers $a$ and $b$ such that $a \leq \alpha \leq b$ and $b - a \leq 10^{-3}$. Justify the choice of the initial interval.
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with abscissa 0. Is the position of the curve relative to this single tangent enough to state that $f$ is concave on its whole domain? Justify.